MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH AND SCIENCE

School of Computer Application

LAB PRACTICAL FILE

Data Structure and Algorithms

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Semester 1

MCA 1A

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EXPERIMENT 1

AIM : To traverse an array, insertion of a new element, deletion of an element into array.

Output 1st :

```
#to traverse
a = [10,20,30,40,
for i in range (]
```

Input 1st :

```
10
20
30
40
```

Output 2nd :

```
#To insert a new
a = [10,20,30,40,
a.insert(5,60)
```

Input 2nd :

```
[10, 20, 30, 40, 50, 60]
PS C:\Users\shikh\OneDrive\D
```

Output 3rd :

```
#To delete an el
a = [10,20,30,40
a.remove(40)
```

Input 3rd :

```
[10, 20, 30, 50]
PS C:\Users\shikh\OneDrive\Des
```

EXPERIMENT 2

Aim : Find Product of 2 matrices

Output :

```
#Product of 2 matrices
a = [[2,4],
     [1,5]]
b = [[3,6],
     [2,1]]
result = [[0,0],
          [0,0]]
for i in range(2):
```

Input :

```
[[14, 16], [13, 11]]
PS C:\Users\shikh\OneDrive\Des
```

EXPERIMENT 3

Aim : To find Electricity bill units

Output :

```
#Program1 Electricity bills
units = int(input("enter unit
if units>=1 and units<=100:
    print("price:",units*4)
elif units>=101 and units<=30
    print("price:",units*6)
```

Input :

```
enter units cons
price: 2600
```

EXPERIMENT 4

Aim : To Find largest of three numbers and smallest of three numbers.

Output 1st :

```
#Program2 which no. is
a = int(input("enter a
b = int(input("enter b
c = int(input("enter c
if a>b and a>c:
    print("greatest=",
elif b>a and b<c:
```

Input 1st :

```
enter a:45
enter b:24
enter c:67
```

Output 2nd :

```
#Program3 which no. is
a = int(input("enter a
b = int(input("enter b
c = int(input("enter c
if a<b and a<c:
    print("smallest=",
elif b<a and b<c:
```

Input 2nd :

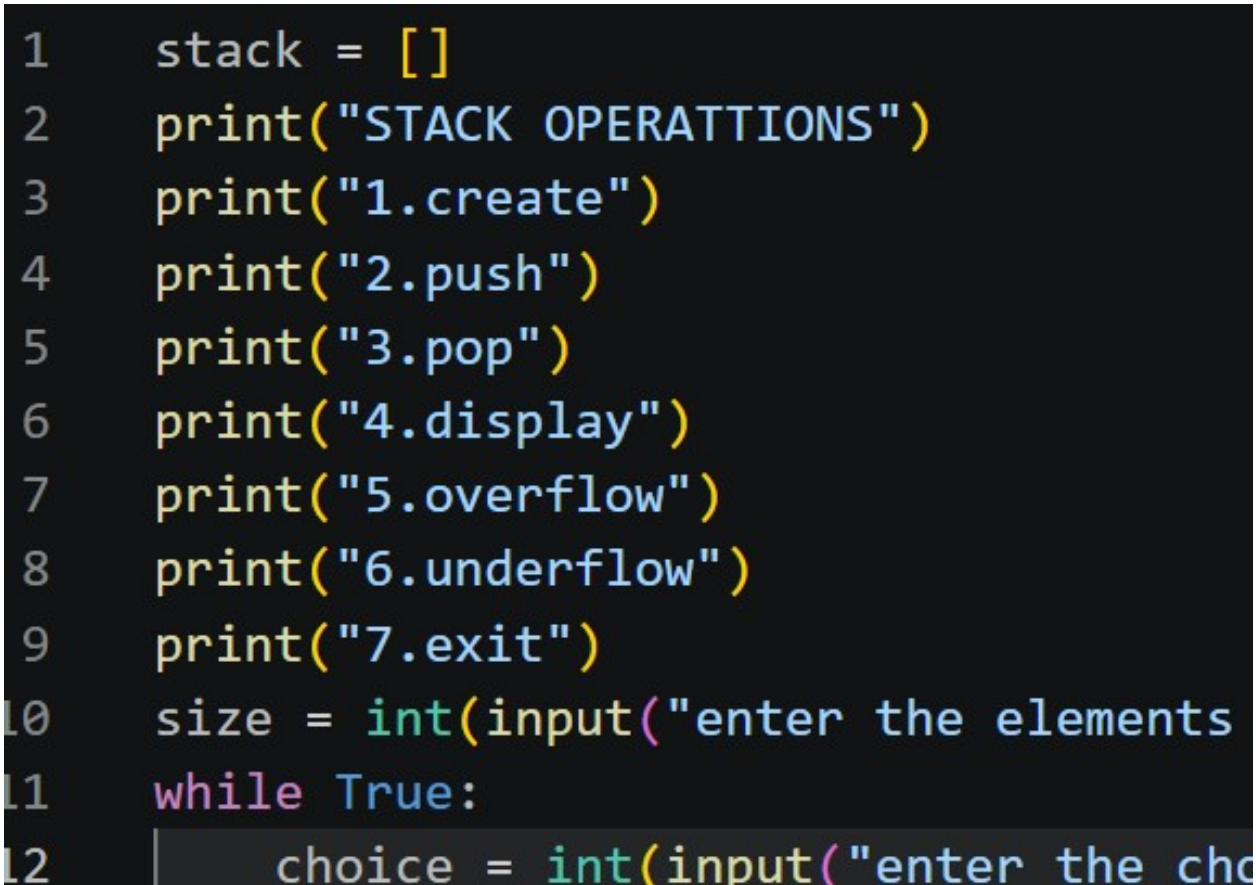
```
enter a37
enter b17
enter c88
smallest= 17
```

EXPERIMENT 5

AIM : Stack Operations-

1. CREATE
2. PUSH
3. POP
4. DISPLAY
5. OVERFLOW
6. UNDERFLOW
7. EXIT

Output :



```
21          if len(stack) == size:
22              print("stack overflow")
23          else:
24              value = int(input("enter  th
25              stack.append(value)
26              print(stack,"value pushed in
27
28      #pop
29      elif choice == 3:
30          if len(stack) == 0:
31              print("stack is underflow")
32          else:
33              value = stack.pop()
41              print("stack elements ar
42              for i in range(len(stack
43                  print(stack[i])
44
45      #overflow
46      elif choice == 5:
47          if len(stack) == size:
48              print("stack is in overf
49          else:
50              print("stack is not over
51
52      #underflow
```

```
59      #exit
60      elif choice == 7:
61          print("program ended")
62          break
```

Input :

```
STACK OPERATTIONS
1.create
2.push
3.pop
4.display
5.overflow
6.underflow
7.exit
enter the elements : 10
enter the choice : 2
enter  the element to ad
[20] value pushed into s
enter the choice : 3
20 poped from stack
enter the choice : 4
```

EXPERIMENT 6

Aim : To check whether the parentheses in a given expression are balanced or not using a stack.

Output :

```
exp = input("Enter an expressio
stack = []
balanced = True

for ch in exp:
    if ch in "([{":
        stack.append(ch)

    elif ch in ")]}":
        if not stack:
            balanced = False
            break


if stack:
    balanced = False

if balanced:
    print("Parentheses are bala
```

Input :

